



LaSalle County Station
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10 CFR 50.73

RA17-064

July 14, 2017

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2017-006-00, Low Pressure Core Spray
Inoperable due to Minimum Flow Valve Failure in Closed Position

In accordance with 10 CFR 50.73(a)(2)(v)(D), Exelon Generation Company, LLC (EGC) is submitting Licensee Event Report (LER) Number 2017-006-00 for LaSalle County Station, Unit 1.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Guy V. Ford, Jr., Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "Harold T. Vinyard", written over a horizontal line.

Harold T. Vinyard
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station

**LICENSEE EVENT REPORT (LER)**
(See Page 2 for required number of digits/characters for each block)(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

LaSalle County Station, Unit 1

2. DOCKET NUMBER

05000373

3. PAGE

1 OF 3

4. TITLE

Low Pressure Core Spray Inoperable due to Minimum Flow Valve Failure in Closed Position

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
05	17	2017	2017	- 006	- 00	07	17	2017	NA	NA
9. OPERATING MODE			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)	☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		
			☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		
			☐ 20.2203(a)(1)	☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		
			☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		
10. POWER LEVEL			☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)		
			☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)		
			☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)		
			☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)		
			☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)		
			☐ 50.73(a)(2)(i)(C)		☐ OTHER		Specify in Abstract below or in NRC Form 366A			

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

John Keenan, Engineering Director

TELEPHONE NUMBER (Include Area Code)

(815) 415-3800

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	BM	FS	S382	Y	NA	NA	NA	NA	NA

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR
NA	NA	NA

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On May 17, 2017 at 0908 CDT, during Unit 1 full-power operations, operators received an unexpected alarm for the Low Pressure Core Spray (LPCS) pump injection high flow and automatic closure of the LPCS minimum flow valve (1E21-F011). Inspections indicated the flow switch that actively controls the LPCS minimum flow valve had a faulty diaphragm which allowed for water intrusion into the device. There were no impacts on plant operations. The required actions of Technical Specifications 3.5.1, "ECCS - Operating" and TS 3.3.5.1, "Emergency Core Cooling System (ECCS) Instrumentation" were entered. The switch was replaced and LPCS system tested, which allowed full restoration of the system on May 17, 2017 at 18:45 CDT.

This condition could have prevented the LPCS system from performing its design function. This condition is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. There was minimal safety consequences associated with the condition since other emergency safety systems remained operable.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2017	- 006	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

LaSalle County Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power.

The Low Pressure Core Spray (LPCS) system is part of the Division 1 Emergency Core Cooling System (ECCS). LPCS provides low-pressure coolant into the reactor core under accident conditions. The LPCS minimum flow switch 1E21-N004 monitors pump discharge flow. This flow switch initiates opening of LPCS minimum flow valve 1E21-F011, if flow drops below the set point (1325 gpm) and initiates closing of 1E21-F011, if flow rises above its reset point (1774 gpm).

CONDITION PRIOR TO EVENT

Unit(s): 1	Event Date: May 17, 2017	Event Time: 0908 CDT
Reactor Mode(s): 1	Mode(s) Name: Power Operation	Power Level: 100 percent

DESCRIPTION OF EVENT

At 0908 CDT on May 17, 2017, the Unit 1 Main Control Room received an unexpected alarm on panel 1H13-P601-C508 for the LPCS pump injection high flow and automatic closure of the LPCS minimum flow valve 1E21-F011. The automatic closure of the LPCS minimum flow valve occurred as appropriate for this alarm condition. An operator was dispatched, who verified that there were no abnormal conditions in the LPCS pump room and that there were no abnormal indications at the 1E21-N004 flow switch, which is the device that initiates this alarm and subsequent actions. LPCS was declared inoperable and the appropriate Technical Specification (TS) required actions were entered in accordance with TS 3.5.1, "ECCS – Operating" and TS 3.3.5.1, "Emergency Core Cooling System (ECCS) Instrumentation." There were no related work activities in progress at the time the condition was identified.

The applicable TS required action (RA) entries were TS 3.5.1 RA A.1, restore LPCS to operable in 7 days; TS 3.3.5.1 RA A.1 to enter condition referenced in Table 3.3.5.1-1 for the LPCS pump discharge flow-low (bypass) channel; TS 3.3.5.1 RA D.1 to declare supported features inoperable when its redundant feature ECCS initiation capability is inoperable; and, TS 3.3.5.1 RA D.4 to restore the channel to operable in 7 days.

Instrument maintenance technicians inspected the LPCS minimum flow switch and found water intrusion in the switch electrical housing. This switch was removed and sent offsite for failure analysis. The failed switch was replaced and calibrated. Unit 1 LPCS was declared operable on May 17, 2017 at 18:45 CDT.

CAUSE OF EVENT

Water intrusion into the switch electrical housing for the LPCS minimum flow switch 1E21-N004 resulted in a ground fault causing the actuation of relay 1E21-K006 and subsequent LPCS minimum flow valve closure. The flow switch was replaced with a like for like replacement, and the switch was sent to Power Labs for failure analysis. The cause was determined to be process water leakage past an internal "U-cup" seal around the shaft from the sensing element into the electrical switch housing. This failure has previously occurred with a similar model of flow switch used in another system's application.

The causal investigation concluded the design of this component was not optimal for its function. This was the first failure of this type noted with model 141N6-WX7-M9 that is utilized for residual heat removal (RHR) and LPCS minimum flow applications. A previous design change to improve the reliability of static O-ring (SOR) differential pressure switches and associated failure analysis did not evaluate all SOR switches using the deficient U-cup design.

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NARRATIVE**REPORTABILITY AND SAFETY ANALYSIS**

This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that condition that could have prevented fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident. An Emergency Notification System (ENS) report number 52761 was made on May 17, 2017 at 12:29 EDT.

The safety significance of this condition was minimal, as plant equipment responded as expected for the event; and all plant parameters remained within plant safety or design limits. Additionally, other emergency safety systems remained operable and there were no related work activities in progress at the time the condition was identified.

The condition is a safety system functional failure (SSFF) defined in accordance with NEI 99-02, Regulatory Assessment Performance Indicator Guidance. The failure of the 1E21-N004 switch caused the minimum flow valve to close when not demanded. Although this did not prevent the LPCS pump from injecting into the vessel to perform its safety function, the conditions of the system upon an automatic start could cause the LPCS pump to run without a discharge path, causing failure of the pump due to a dead-head condition.

CORRECTIVE ACTIONS

The LPCS minimum flow switch 1E21-N004 was replaced and LPCS system tested, which allowed full restoration of the system. Investigation of the failure determined that cause was due to design of the U-cup used for this application. Additional corrective actions are to schedule the replacement of the current design of SOR differential pressure switches in other systems, as part of an extent of condition review, and to evaluate the failure analysis for additional actions as needed.

PREVIOUS OCCURRENCES

A review of station Licensee Event Reports in the past three years did not identify a similar occurrence. However, searches were also performed of the station's corrective action program (CAP) and INPO operating experience databases, which found the following similar occurrences.

- ICES 308842, Unexpected Isolation of Reactor Core Isolation Cooling (RCIC) System
On June 13, 2013, a RCIC high steam flow differential pressure switch (SOR model 141N6-WX47-M9-C1A-JJTNNQ) was found to be in the tripped condition. When the leads to the switch were lifted, the relay causing the isolation signal dropped out. Upon further investigation, the switch was identified as having internal corrosion. It was determined that water leaked into the switch housing through a U-cup style seal and over time caused rust to build up on the switch internals eventually resulting in a false actuation and the isolation of RCIC. The switch was replaced, and the vendor developed a new design of the Model 141N6 switch that uses an O-ring rather than the U-cup seal.
- ICES 312061, Failure of Differential Pressure Switch Due to Crack in Diaphragm
On May 27, 2014, during an instrument surveillance, the equalizer valve for a SOR model 131N6-BX5-M9-C1A-JJTNNQ differential pressure switch (2E31-N010C Main Steam "C" High Flow Detection) was isolated. Leak-by of the equalizer valve was noted, which prevented completion of the calibration. This event necessitated switch replacement. Subsequent failure analysis identified that the leak-by was the result of a circumferential crack in the diaphragm in the switch. Due to redundancy in the system, this issue had no impact on system reliability or unit operation.

COMPONENT FAILURE DATA

Manufacturer: Static O-Rink SOR Inc. (S382)
Device (Model): Differential Pressure Switch (141N6-WX7-M9-C1A-JJTNNQ)
Component ID: LPC Pump Discharge Flow Bypass Interlock (1E21-N004)